

Addition and subtraction of radicals



1 Simplify the following:

a $8\sqrt{3} + 18\sqrt{3}$

c $8\sqrt{6} + 17\sqrt{13} + 19\sqrt{6} - 6\sqrt{13}$

e $\sqrt{6} + \sqrt{54}$

g $\sqrt{32} - \sqrt{98}$

i $\sqrt{10} + 10\sqrt{10}$

k $18\sqrt{2} - 10\sqrt{2} + 20\sqrt{2}$

m $\sqrt{243} + \sqrt{3}$

o $\frac{12 + 9\sqrt{5}}{3}$

b $-12\sqrt{5} + 6\sqrt{5} - 12\sqrt{5}$

d $17\sqrt{6} - 8\sqrt{11} + 3\sqrt{6} + 13\sqrt{11}$

f $\sqrt{180} + \sqrt{500}$

h $4\sqrt{45} + 5\sqrt{180}$

j $-19\sqrt{5} + 4\sqrt{5}$

l $19\sqrt{3} + 19\sqrt{7} - 7\sqrt{7} + 22\sqrt{3}$

n $2\sqrt{75} - 3\sqrt{108}$

2 Use the formula $\sqrt{a} + \sqrt{b} = \sqrt{a + b + 2\sqrt{ab}}$ to simplify the following:

a $\sqrt{3} + \sqrt{12}$

b $\sqrt{8} + \sqrt{2}$

c $\sqrt{18} + \sqrt{8}$

d $2\sqrt{20} + 2\sqrt{5}$

3 Simplify the following:

a $\sqrt[3]{512v} - 5\sqrt[3]{v}$

b $2\sqrt[3]{u} + \sqrt[3]{125u}$

c $\sqrt{ax^5} + x^2\sqrt{ax}$

Multiplication and division of radicals

4 Simplify the following:

a $\sqrt{2} \times \sqrt{11}$

b $2\sqrt{2} \times 2$

c $4\sqrt{35} \times \sqrt{5}$

d $5\sqrt{7} \times 3\sqrt{5}$

e $\sqrt{5} \times \sqrt{2} \times \sqrt{7}$

f $\sqrt{343} \times \sqrt{75}$

g $\sqrt{5} \times \sqrt{7}$

h $8 \times 10\sqrt{5}$

i $\sqrt{7} \times \sqrt{3} \times \sqrt{11}$

j $\sqrt{55} \times \sqrt{11}$

k $4\sqrt{11} \times 5$

l $2\sqrt{5} \times 15\sqrt{11}$

m $7\sqrt{22} \times \sqrt{2}$

n $\sqrt{180} \times \sqrt{48}$

o $8\sqrt{15} \times 8\sqrt{5}$

p $5\sqrt{17} \times 8\sqrt{3}$

q $17\sqrt{35} \times 4\sqrt{5}$

r $8\sqrt{51} \times 9\sqrt{3}$

5 Simplify the following:

a $\sqrt{85} \div \sqrt{17}$

b $18\sqrt{7} \div 9$

c $24\sqrt{112} \div 3\sqrt{7}$

d $8\sqrt{77} \div \sqrt{7}$

e $20\sqrt{14} \div 5\sqrt{2}$

f $\sqrt{15} \div \sqrt{5}$

g $\sqrt{55} \div \sqrt{5}$

h $\sqrt{51} \div \sqrt{17}$

i $\sqrt{21} \div \sqrt{3}$

j $\sqrt{91} \div \sqrt{7}$

k $40\sqrt{7} \div 8$

l $10\sqrt{55} \div \sqrt{11}$

m $15\sqrt{22} \div \sqrt{11}$

n $4\sqrt{35} \div 2\sqrt{5}$

o $\sqrt{27} \div \sqrt{3}$

p $3\sqrt{20} \div \sqrt{5}$

q $5\sqrt{8} \div \sqrt{2}$

r $40\sqrt{96} \div 10\sqrt{6}$

s $50\sqrt{24} \div 10\sqrt{6}$

t $\sqrt{25} \div \sqrt{81}$



6 Simplify the following:

a $\sqrt{\frac{28}{7}}$

b $\sqrt{\frac{9}{45}}$

c $\sqrt{\frac{64}{4}}$

d $\sqrt{\frac{48}{144}}$

e $\frac{\sqrt{12}}{\sqrt{36}}$

f $\frac{\sqrt{56}}{\sqrt{14}}$

g $\frac{\sqrt{36}}{\sqrt{81}}$

h $\frac{\sqrt{72}}{\sqrt{32}}$

i $\frac{17\sqrt{21}}{\sqrt{3}}$

7 Assuming that all variables are non-negative, simplify the following:

a $\sqrt{4j}\sqrt{j}\sqrt{4k}$

b $y\sqrt{y}y\sqrt{y}$

8 Simplify the following:

a $\frac{\sqrt[3]{6}}{\sqrt[3]{6} \times 7^3}$

b $\frac{\sqrt{2p}}{\sqrt{2q}}$

c $6\sqrt{7m} \div \sqrt{4m}$

Mixed operations with radicals

9 Distribute and simplify the following:

a $\sqrt{7}(-5 + \sqrt{5})$

b $7\sqrt{2}(\sqrt{7} + 6)$

c $7\sqrt{11}(-6\sqrt{2} + \sqrt{7})$

d $\sqrt{11}(\sqrt{7} + 4)$

e $\sqrt{7}(3 + \sqrt{3})$

f $\sqrt{2}(\sqrt{11} - 6)$

g $3\sqrt{3}(\sqrt{13} - 5)$

h $\sqrt{3}(\sqrt{11} + \sqrt{13})$

i $4\sqrt{7}(\sqrt{2} - \sqrt{11})$

j $3\sqrt{5}(\sqrt{55} + \sqrt{11})$

k $8\sqrt{2}(\sqrt{3} - 3\sqrt{7})$

l $5\sqrt{2}(3\sqrt{5} + 4\sqrt{7})$

m $7\sqrt{3}(\sqrt{15} + \sqrt{60})$

n $11\sqrt{3}(3\sqrt{5} - \sqrt{20})$

o $8\sqrt{11}(3\sqrt{7} - 4\sqrt{5})$

p $\sqrt{13}(\sqrt{11} + 2)$

q $8\sqrt{11}(-3\sqrt{5} - 7\sqrt{7})$

10 Distribute and simplify the following:

a $(\sqrt{13} + 5)(\sqrt{2} - 3)$

b $(\sqrt{7} - \sqrt{13})(\sqrt{2} - \sqrt{11})$

c $(\sqrt{11} + 1)^2$

d $(2\sqrt{5} + 3)^2$

e $(5 - \sqrt{7})(8 - \sqrt{3})$

f $(3\sqrt{11} - \sqrt{5})^2$

g $(\sqrt{11} - 11)(\sqrt{11} + 11)$

h $(\sqrt{10} + \sqrt{7})(\sqrt{10} - \sqrt{7})$



11 Simplify the following:

a $3\sqrt{5} \times 8\sqrt{10} \div 2\sqrt{2}$

b $\frac{8\sqrt{12} \times 3\sqrt{2}}{4\sqrt{10}}$

c $\frac{\sqrt{15} + \sqrt{21}}{\sqrt{3}}$

d $\frac{3\sqrt{2} \times 2\sqrt{6}}{\sqrt{8} \times \sqrt{20}}$

12 Find the values of x and y in the following equations:

a $(8\sqrt{91} - 5\sqrt{7})^2 = x + y\sqrt{13}$

b $(\sqrt{x} + y)^2 = 14 - 6\sqrt{5}$

13 Consider the expression $(\sqrt{13} + 4)^2 + (\sqrt{13} + m)^2$.

a Distribute and simplify the expression.

b What value of m can be substituted into $(\sqrt{13} + 4)^2 + (\sqrt{13} + m)^2$ so that it has a rational value?

14 Simplify the following:

a $\sqrt[3]{u^2} \sqrt[3]{u}$

b $\sqrt[3]{3u^2} \sqrt[3]{9u}$

c $7\sqrt{u} + 3\sqrt{u}$

d $10\sqrt{u} - 8\sqrt{v} - 4\sqrt{u} + 9\sqrt{v}$

e $6\sqrt{a} + \sqrt{81a}$

f $\sqrt{125a} - \sqrt{20a}$

Rationalize the denominator

15 State whether the following results in a rational number:

a $(\sqrt{8} + \sqrt{3}) \times (\sqrt{8} + \sqrt{3})$

b $(\sqrt{8} + \sqrt{3}) \times (\sqrt{8} - \sqrt{3})$

c $(-(\sqrt{8} + 2)) \times \sqrt{8}$

16 Rationalize the denominator of the following:

a $\frac{1}{\sqrt{11}}$

b $\frac{\sqrt{2} - 2}{\sqrt{11}}$

c $\frac{7}{\sqrt{12}}$

d $\frac{14\sqrt{11}}{9\sqrt{13}}$

e $-\frac{3}{5\sqrt{13}}$

f $\frac{7\sqrt{30}}{\sqrt{10}}$

g $\frac{10\sqrt{7} - 1}{\sqrt{5}}$

h $\frac{\sqrt{5}}{\sqrt{15}}$

17 Find a conjugate for the following expressions:

a $2 + \sqrt{7}$

b $9 - \sqrt{11}$

c $4 - 2\sqrt{10}$

d $5\sqrt{2} + 4\sqrt{10}$



18 Rationalize the denominator for the following expressions:

a $\frac{9}{\sqrt{5} - 7}$

b $\frac{28}{\sqrt{10} - \sqrt{3}}$

c $\frac{\sqrt{7}}{5\sqrt{6} + 4\sqrt{2}}$

d $\frac{\sqrt{11} + \sqrt{2}}{\sqrt{11} - \sqrt{2}}$

e $\frac{9\sqrt{7} + 8\sqrt{5}}{9\sqrt{7} - 8\sqrt{5}}$

f $\frac{8}{9\sqrt{11} - 5}$

g $\frac{\sqrt{7} + \sqrt{2}}{\sqrt{7} - \sqrt{2}}$

h $(\sqrt{10} + 1)^{-1}$

19 Express the following in simplest form with rational denominator:

a $\frac{3}{\sqrt{10}} + \frac{1}{4}$

b $\frac{3}{4\sqrt{5}} - \frac{3}{2\sqrt{2}}$

c $\frac{5\sqrt{3}}{6\sqrt{5}} - \frac{7\sqrt{5}}{3\sqrt{3}}$

d $\frac{1}{\sqrt{2} + 5} + \frac{1}{\sqrt{2} - 5}$

20 Show that the following expression is a rational number:

$$\frac{3}{\sqrt{2} + 3} + \frac{3}{\sqrt{2} - 3}$$

21 If $x = 2 + \sqrt{3}$, simplify and rationalize the denominator of the following expressions:

a $\frac{x - \sqrt{3}}{x + \sqrt{3}}$

b $\left(x + \frac{1}{x}\right)^2$

22 If $x = 4 + \sqrt{3}$, simplify and rationalize the denominator of the following expressions:

a x^2

b $\frac{x + 1}{x - 1}$

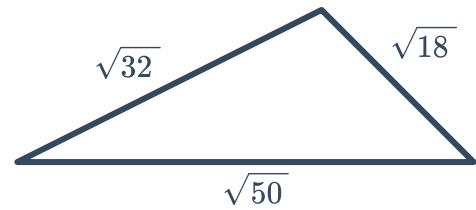
23 Express the following as a single radical:

$$42 + \frac{6}{7 + 5\sqrt{2}}$$

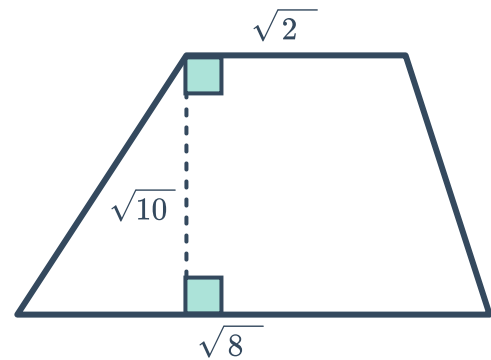


Applications

- 24** Find the perimeter of the triangle in simplified radical form:



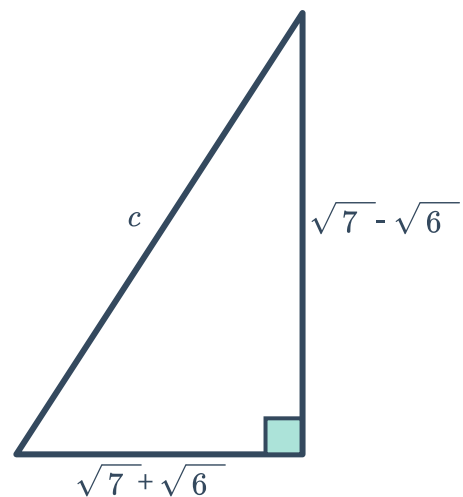
- 25** Find the area of the trapezoid in radical form:



- 26** A rectangle has a width of $(\sqrt{7} - 2)$ ft and a length of $(\sqrt{7} + 2)$ ft.

- a** Find the exact perimeter of the rectangle.
- b** Find the exact area of the rectangle.

- 27** Find c , the length of the hypotenuse of the triangle below. Leave your answer in radical form.





28 Find the exact perpendicular height of a triangle whose area is $36\sqrt{15}$ square centimeters and whose base measures $8\sqrt{5}$ centimeters.

29 A rectangle with height h and width w is believed to be more aesthetically pleasing if its dimensions are in the following ratio:

$$\frac{w}{h} = \frac{2}{\sqrt{5} - 1}$$

- a** Rationalize the denominator of the ratio.
- b** Find a decimal approximation for this ratio correct to the nearest thousandth.

30 A rectangle has a length of $(\sqrt{10} + 3)$ meters and an area of 4 square meters. What is the exact width of this rectangle? Give your answer with a rational denominator.

31 The formula $P = \frac{x(12 + \sqrt{x})}{5\sqrt{x}}$ models the percentage, P , of bees that do not survive relocation to a new hive in a colony that has x thousand bees.

- a** In a hive with 25 000 bees, what percentage will not survive relocating to a new hive? Round your answer to the nearest integer.
- b** Rewrite the formula for P by rationalizing the denominator.
- c** Using the formula you found in part (b), find the percentage of bees from a hive with 25 000 bees that will not survive relocation. Round your answer to the nearest integer.

32 Consider a square pyramid with a height h and a volume V . The length of one side of its base is given by $\sqrt{\frac{3V}{h}}$.

- a** Rewrite the expression for the length such that it has a rational denominator.
- b** Find the side length of the base of a square pyramid that has a height of 3 inches and a volume of 16 cubic inches.